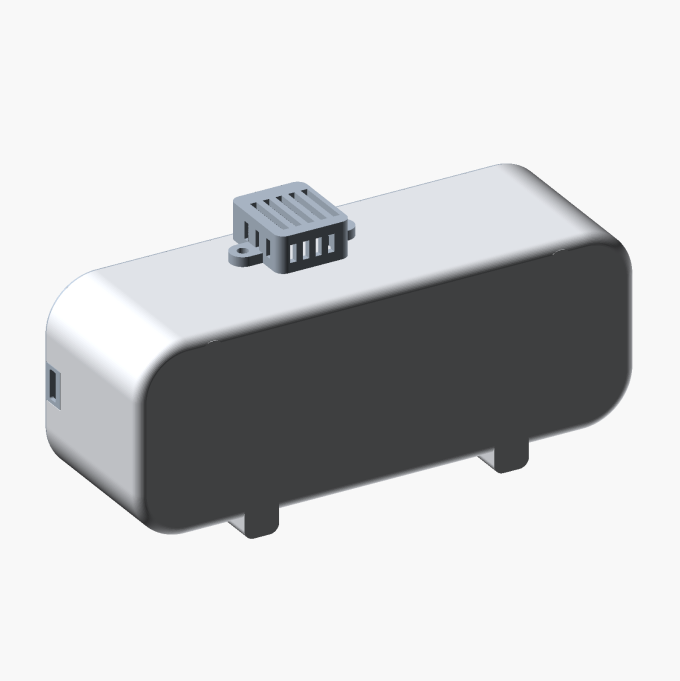
ESP32 Smart Clock

ENCLOSURE / PRINT & ASSEMBLY GUIDE / REVISION 18



Three printed parts. One rounded enclosure.

Print one **body.stl**, one **lid.stl** and one **sensor_cage.stl**. All files use millimetres and must stay at 100% scale. The two small fit samples are optional checks, not assembly parts.

Dimension	Value
Body	158 x 60 x 49.7 mm
Overall with lid, feet and sensor cage	158 x 77.5 x 51.2 mm
Side walls / lid / optical front	2.25 mm / 1.5 mm / 0.6 mm

Mandatory lid pause: fixed 0.15 mm layers, including the first. Pause after **Z 4.05 mm**, before Z 4.20 mm, to insert two M3 nuts. See page 3 before slicing.

The creator tested the 0.6 mm front diffuser successfully in white PLA. The complete latest assembly and insert-pause process are geometrically checked but not physically validated. Images in this guide are CAD renders.

Print setup & hardware

Part	Material / orientation	Layers
Body	White PLA; front face on bed, as supplied. At least three solid bottom layers through the diffuser.	0.20 mm first and later
Lid / ESP sample	PLA; outer face on bed, as supplied. Mandatory pause on page 3.	0.15 mm first and later, fixed
Sensor cage / power sample	PLA; supplied orientation. Inspect bridges and small openings.	0.20 mm first and later

Design target: Ender-3 V3 KE. Suggested starting point: 0.4 mm nozzle, three perimeters, 20% infill. Use calibrated temperatures and speeds. Inspect bridges and overhangs, keeping supports out of the insert pockets, vents and wire routes. No validated slicing profile or G-code is supplied. Opaque black PLA blocks the display.

Fasteners

14 M3 brass inserts: 2.9 mm long, maximum OD 4.2 mm (root about 3.7 mm). Pilot holes are nominally 3.9 mm. Flat-underhead screws: measured head OD 5.2 mm, height 2.1 mm.

Location	Count	Screw / washer
Lid to body	4	M3 x 5 + 0.5 mm washer, or M3 x 4 without washer
Sensor cage	2	M3 x 6
ESP at USB side	2	M3 x 8 + 0.5 mm washer; inserts fitted mid-print
Other ESP posts	2	M3 x 5
Speaker	2	M3 x 5 + 0.5 mm washer
Amplifier	2	M3 x 5 + 0.5 mm washer, washer OD 7 mm

Make sure each screw clamps its part before reaching the bottom of its hole. The ESP PCB thickness was inferred as 2.5 mm; confirm against your board. USB-side screw tips have about 0.9 mm nominal clearance from the closing skin.

Test before printing the full parts

Use **esp_gauge.stl** to check the two relieved ESP posts, button clearance and pause workflow. Use **power_gauge.stl** to check the 13.1 x 5.5 mm oblong opening with the actual snap-in socket. Do not alter model scale to correct a local fit issue.

Lid: install two inserts mid-print

The two rear access holes have built-in 0.6 mm closures. After the lid is complete, the narrower screw passages prevent installing these two inserts from above.

Feature	Height from print bed
Closed outside skin	0.00 to 0.60 mm
Insert supporting shoulder	1.05 mm
Top of seated 2.9 mm insert	3.95 mm
Pause after completed layer	4.05 mm (27 completed layers)
Next layer after resume	4.20 mm

1. Keep the supplied lid orientation, with the outside face on the bed. Set 0.15 mm for the first and all later layers. Do not add a raft or use adaptive layers.

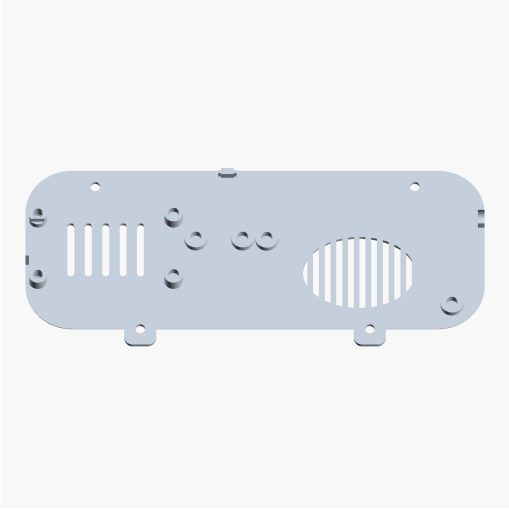
2. Set the pause after Z 4.05 mm and before Z 4.20 mm. Verify height in the slicer preview; some slicers number layers differently. Use the pause action supported by your printer.

3. At the pause, heat-set two M3 inserts into the open USB-side pockets. The 3.9 mm pilot is smaller than the 4.2 mm brass insert. Do not force them in cold or remove the print from the bed.

4. Seat each insert on its internal shoulder. Its top should sit at Z 3.95 mm, about 0.10 mm below the paused surface. Confirm both are below the nozzle path.

5. Resume printing. The upper post sections capture the inserts. Fit other accessible inserts after printing and cooling. Test this process on the small ESP sample first.

Lid mounting face



Fit & assemble

Measured hardware	Envelope / key fit
MAX7219 display	132.1 x 32.1 x 17.8 mm including wires
ESP32 USB-C board	51.6 x 28.2 mm PCB; 53.5 mm including USB; 23 mm wired depth; hole centers 2.45 mm from adjacent edges
Speaker	69.7 x 31.2 x 16.6 mm; 3.3 mm holes; 1.7 mm mounting ears
NS4168 amplifier	38.5 x 22.4 mm PCB; 42.4 mm wired length; 10.1 mm height
External sensor	15 x 15 x 7.6 mm with connector; loose in cage
Power USB-C	Body 12.8 x 5.2 mm; depth 12.7 mm behind flange; flange 15.6 x 9.3 mm; spring span up to 15.7 mm

1. Let the body cool before release; do not push on the thin front. Fit the lid and cage inserts into the body.

2. Seat the display against the diffuser between the four L-shaped stops. No display screws or snap arms are used.

3. Mount ESP, amplifier and speaker on the lid. ESP USB faces the left port. The speaker membrane faces the rear grille; speaker wires lead toward the amplifier.

4. With the lid open, insert the power socket into the opposite side opening near the speaker. Its long axis follows the depth of the case.

5. Route the connected wiring. Feed sensor wires through the body opening within the cage footprint. Close the lid without pinching wires, then fit four screws.

6. Rest the sensor on the body outside and secure the ventilated cage with two opposite screws. Do not pass M3 screws through the sensor PCB.

Validation & reuse

All assembly meshes and fit samples are single watertight solids. Tests cover rigid component envelopes, insertion clearances, vents and connector geometry. Measurement uncertainty is about +/-1 mm; actual button shapes, spring deflection and wiring still need physical fitting. Nominal display-to-ESP wiring clearance is 5.1 mm.

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github.com/NirBY/ESP32_Smart_Clock
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